



OPEN Spatiotemporal differentiation and influencing factors of the degree of resilience coupling coordination in the Beijing–Tianjin–Hebei region, China

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Urban agglomeration system resilience involves multiple subsystems that are interconnected and interactive, and exploring the relationships among subsystems can elucidate and increase urban agglomeration resilience. This study constructed a social-economic-built environment resilience model and proposed a social-economic-built environment resilience index system. Using the entropy weight method, the coupling coordination degree (CCD) model, GeoDetector, spatial autocorrelation analysis and the autoregressive integrated moving average model (ARIMA) model, the resilience development level of the Beijing–Tianjin–Hebei (BTH) region from 2000 to 2020 was evaluated, the main influencing factors for the spatiotemporal differentiation pattern were identified, and the development levels in 2030 and 2050 were predicted. The results show that (1) high-resilience and high-CCD regions are concentrated in the region of Beijing and Tianjin in the centre, and low-value regions are concentrated in the districts and counties in the west and south; (2) economic elements, especially the economic aggregate status of a region as a whole, are the main factors contributing to the differences in resilience levels between regions, but their influence gradually decreases; and (3) the gap between the CCDs of Beijing, Tianjin and cities in Hebei increased from 2000 to 2020, but this gap is expected to narrow by 2030 and 2050. On this basis, five categories of future development guidelines for cities in the BTH region are proposed.

Keywords Urban agglomeration resilience, Coupling coordination degree, Spatiotemporal evolution, Influencing factors, Beijing–Tianjin–Hebei Region

Resilience refers to the ability of a system or society to remain stable, adapt and recover in the face of external shocks and internal stresses¹. Resilience is the ability to respond to natural disasters, economic crises, social change, etc.^{2,3}. The concept of resilience shifts the focus from the disaster to the ability of the disaster-bearing body to cope with the disaster, emphasizing the importance of keeping losses within acceptable limits by increasing the resilience of the object, rather than rigidly preventing the disaster from occurring (which is usually impossible). This concept provides a new, actionable perspective on safety and disaster prevention. In particular, the concept of urban resilience encompasses many interrelated aspects, including adaptability, resistance and recovery. Adaptability can be defined as the accumulated experience of a city in the context of previous disasters. Resistance denotes the capacity of a city to utilize its inherent capabilities to withstand disasters and maintain operational continuity. Recovery is the ability to facilitate a city's return to its original state or a new equilibrium following a disaster⁴. Resilience can be further categorized into urban agglomeration resilience, city resilience, community resilience, etc., on the basis of the different scales of the subject⁵. As a composite, interconnected and diverse system, the resilience of urban agglomerations has unique characteristics compared with the resilience at other scales. There are strong economic, social and infrastructural interdependencies between cities within clusters⁶. This interdependence means that the resilience of urban areas depends not only on the capacity of individual systems, but also on the coordination and cooperation of the whole network^{7,8}. The resilience characteristic of urban agglomerations reflects their complexity and dynamism as systems of multiple cities

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and communities that interact, requiring a combination of social, economic, environmental and technological factors to be considered⁹.

To quantify the resilience of urban agglomerations to better understand the complex operational relationships within the system, researchers have conducted quantitative assessment studies on the resilience of urban agglomerations. The existing studies have focused mainly on the construction of an evaluation index system, the theoretical framework and methodology, and the evaluation of the resilience of specific urban agglomerations. With respect to the construction of the evaluation indicator system, studies have been conducted from different perspectives to construct a target framework and a quantitative indicator system that includes several subsystems. For example, the evaluation indicator system is constructed from four dimensions: economic, political, social and infrastructure¹⁰; the network used to assess the resilience of urban areas is constructed in terms of three dimensions of economic, information and transport connectivity¹¹; an ecological resilience evaluation model is constructed from three dimensions of resistance, adaptability and recovery¹²; the urban disaster resilience evaluation index system is constructed from infrastructure, economic, social and ecological aspects¹³; and a regional ecological vulnerability evaluation system is constructed on the basis of the sensitivity–resilience–pressure (SRP) model¹⁴. In the area of theoretical frameworks and methodological research, researchers have combined the complex network approach and spatial econometrics to analyse the impact of economic networks on the economic resilience capacity of cities¹⁵; assessed the level of resilience of urban agglomerations via the TOPSIS method, particle swarm optimization (PSO), extreme learning machines (ELMs), etc.¹⁶; and combined genetic algorithms, integrated entropy methods, kernel density estimation methods, etc., to measure the resilience of urban agglomerations^{17,18}. In the area of resilience assessment for specific urban agglomerations, researchers have conducted studies on the Yangtze River Delta, Chengdu–Chongqing and central Yunnan Urban Agglomeration^{8,19,20}. In addition, studies have been carried out to compare the resilience of several urban agglomerations and to compare and analyse the differences among urban agglomerations⁹. In summary, most existing studies consider multiple systems of urban agglomerations as a whole; although researchers have constructed an indicator system from various aspects, such as social, economic and environmental aspects, they only superimpose these subsystems when the overall resilience level of urban agglomerations is assessed; and there is a lack of research on the interrelationships among these subsystems.

The coupling coordination degree (CCD) model provides a way to explore the interrelationships among subsystems^{21,22}. In physics, coupling refers to the phenomenon of two or more systems that interact and influence each other^{23,24}. When the system or elements within the system enhance each other, it is benign coupling; when they constrain each other, it is malignant coupling²⁵. Coordination refers to the degree of benign coupling between two or more systems and can reflect the coordination of development among different elements²⁶. This study defines the degree of interaction, mutual influence and coordinated development of the three subsystems of social, economic and built environments in the resilience of urban agglomerations as the degree of resilience coupling and coordination of socioeconomic–built environment (S–E–B) in urban agglomerations and constructs a CCD mechanism model and an indicator system to quantitatively measure the coupling relationship and the degree of coordination among the three. Moreover, the elements of the subsystems in the urban agglomeration are constantly developing and changing, so by predicting the future development of the subsystems, we can dynamically simulate the future changes in the CCD to take targeted measures to improve the future CCD of the urban agglomeration²⁷.

This study applies the concept of resilience as a guide and constructs a resilience evaluation index system for urban agglomerations under its framework, which contains three subsystems: social resilience, economic resilience, and built environment resilience. On this basis, the BTH region is used as an empirical research object to assess its resilience level and the distribution characteristics of the CCD in time and space from 2000 to 2020 and to detect the influencing factors of the temporal and spatial distribution characteristics. A mathematical model is applied to predict the CCD in 2030 and 2050. The objectives of this study include the following three aspects: (1) to identify the key resilience elements of multisystem urban agglomerations and construct a socioeconomic–built–environmental resilience indicator system; (2) to assess the resilience level and CCD of the BTH region and, in the spatial dimension, to identify the geographic distribution characteristics of the resilience level and CCD, as well as to determine the main influencing factors of the spatial distribution characteristics; and (3) in the temporal dimension, to analyse the temporal evolution process of the resilience level and CCD, as well as the changes in the dominant factors over time, and predict the future CCD pattern. By analysing the characteristics of temporal and spatial evolution from 2000 to 2020, as well as predicting the spatial distribution of the CCD in 2030 and 2050, the main problems and development proposals for coordinating the resilience coupling of different cities in the BTH region are classified and discussed, with the aim of improving the resilience ability of urban agglomerations to address with future risks.

Materials and methods

Study area

The BTH region (113°27′E–119°50′E, 36°05′–42°40′N) is located in the North China Plain and includes Beijing, Tianjin, and 11 prefecture-level cities in Hebei Province, with a total area of approximately 2.18×10^5 km²²⁸. The BTH region as a whole has a high northwest and low southeast topography, with a land use structure dominated by arable and urban land in the southeastern plains and forests and grasslands in the northwestern mountains²⁹ (Fig. 1).

The BTH region plays an important role in China's economy, culture and politics. Beijing, as the capital, is the centre of politics, culture and high-tech industries; Tianjin is a major port city in the north with significant industrial and commercial influence³⁰. Hebei Province, on the other hand, is dominated by heavy industry and manufacturing³¹. The BTH region is densely populated, with a more complex social structure and a high degree of urbanization. The government has been working to consolidate the region, with the aim of capitalizing on

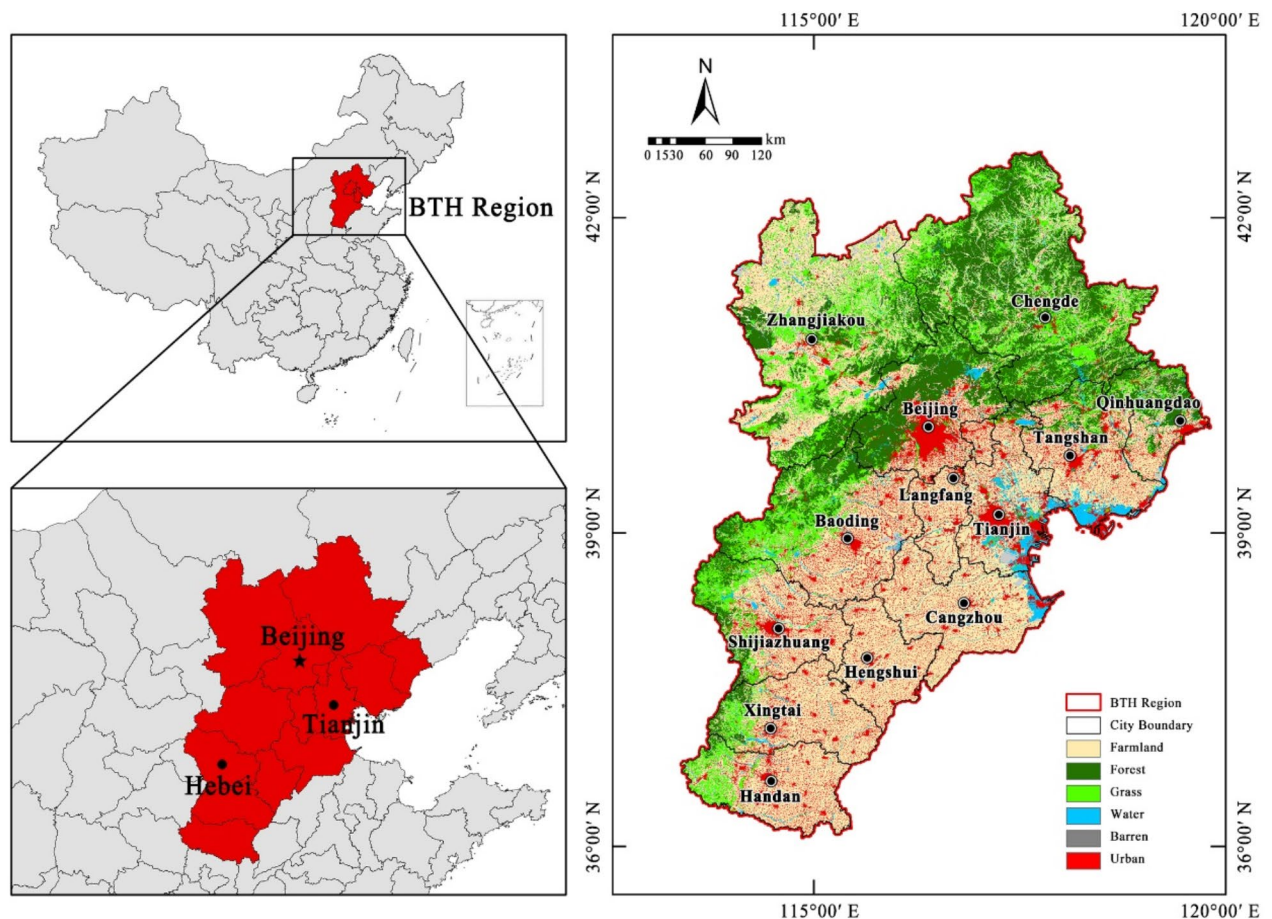


Fig. 1. Overview of BTH region. Produced based on the Ministry of Natural Resources Standard Map Service website, Standard Map No.GS (2016)2923 and No.GS (2016)1610, with no modifications to the boundaries of the base map. The map was created using ArcGIS, version 10.4 (<https://www.esri.com/en-us/arcgis>).

the strengths of each area to create a more efficient and sustainable region³². The cities within the BTH region are considerable in size, and the level of social, economic, and environmental development varies significantly across counties. To circumvent the potential for key information to be overlooked due to homogenization and to facilitate a more detailed analysis, this study employs 199 counties (cities and districts) within the BTH region as the fundamental research unit. This approach is intended to increase the relevance of the findings and facilitate the formulation of more effective policies.

Research methods

Urban agglomeration resilience assessment model

Entropy weighting method The entropy weighting method determines the information value of an indicator according to its degree of dispersion, and the smaller the information entropy is, the greater the degree of dispersion of the indicator, and the greater the influence of the indicator on the overall evaluation (i.e., indicator weights). The entropy weighting method avoids the human bias that may be caused by subjective assignment. Upon the standardization and normalization of the data, this study calculates the weights of the resilience indicators via the entropy weighting method, and the calculation process is as follows:

For each indicator j , its weight in each sample i , p_{ij} is calculated:

$$p_{ij} = Y_{ij} / \sum_{i=1}^n Y_{ij}$$

where n is the number of samples.

Next, the information entropy E_j is calculated for each indicator:

$$E_j = -\ln(n)^{-1} \sum_{i=1}^n p_{ij} \ln p_{ij}$$

When $p_{ij}=0$, $p_{ij} \ln p_{ij}=0$.

The weight of each indicator was determined:

$$W_j = \frac{1 - E_j}{k - \sum E_j}$$

Calculation of the S-E-B resilience index This study applies the linear weighting method to construct the S-E-B resilience index to evaluate the level of resilience under different dimensions, using the social resilience index as an example, and the calculation formula is as follows:

$$SR_j = \sum_{i=1}^n Y_{ij} \times W_j$$

Coupling coordination degree model

Coupling coordination portrays the degree of influence of mutual synergistic effects and roles between two or more systems. By solving the coupling degree C_n between any n systems to indicate the strength of the interaction between the subsystems, which takes a value of 0–1, the formula is as follows:

$$C_n = \left[\frac{U_1 \times U_2 \times \cdots \times U_n}{\left(\frac{U_1 + U_2 + \cdots + U_n}{n} \right)^n} \right]^{\frac{1}{n}}$$

This study involves 3 subsystems; then, n is taken as 3.

Although the degree of coupling can reflect the degree of interaction among social resilience, economic resilience and built environment resilience, it cannot characterize whether the systems are mutually reinforcing at a high level or mutually limiting at a low level. Therefore, the CCD of the composite system is introduced to reflect the nature of the interactions among the systems.

$$D = \sqrt{C \times R}$$

where D is the degree of coupling coordination, the larger its value is, the higher the degree of coordination; C is the degree of coupling; and R is the comprehensive score of the system, which is calculated via the following formula:

$$R = \alpha SR + \beta ER + \gamma BR$$

where α , β , and γ are the weighting coefficients of social resilience, economic resilience and built environment resilience, respectively, and the three systems are considered equally important in this study. Thus, $\alpha = \beta = \gamma = 1/3$.

The uniform distribution function method is used to classify the level of coupling coordination, and the level of CCD of the social, economic and built environment of the BTH region is ranked from low to high (Table 1).

GeoDetector

A geodetector can detect spatial heterogeneity and reveal the driving forces behind it and is used for driving force and factor analysis³³. In this study, we used GeoDetector to measure the relationship between each influencing factor and the system CCD and then identified the dominant factor influencing the CCD, which was calculated via the following formula:

$$q = 1 - \frac{1}{n\sigma^2} \sum_{h=1}^L n_h \sigma_h^2$$

where q represents spatial heterogeneity in the range [0, 1], n represents the total number of samples (n), σ^2 represents the discrete variance, and L represents the number of districts (n).

Local autocorrelation analysis

Local spatial autocorrelation is an indicator for analysing the spatial data of different areas or units in a study area and is used to reflect the degree and significance of spatial differences between each area or unit and its surroundings.

The local Moran index can be used to identify different types of spatial clustering or dispersion phenomena, such as high-value clustered areas, low-value clustered areas, high and low outliers, and low and high outliers that exist in the study area, and its calculation formula is as follows:

$$I_i = \frac{(x_i - \bar{x}) \sum_{j=1}^n w_{ij} (x_j - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

CCD value range	(0.00, 0.25]	(0.25, 0.50]	(0.50, 0.75]	(0.75, 1.00]
CCD grade	Seriously unbalanced	Moderately unbalanced	Basically balanced	Well balanced

Table 1. CCD values and corresponding classification criteria.

where x_i and x_j are the attribute values of the i th and j th elements, respectively, and where w_{ij} is the spatial weight between the i th and j th elements. The clustering and outlier analysis tools in ArcGIS were used to implement this process.

ARIMA forecast

The autoregressive integrated moving average (ARIMA) model was used to predict future changes in the CCD of the resilience index for each unit. The ARIMA model is one method of time series forecasting analysis and can be expressed as follows:

$$Y_t = c + \varphi_1 Y_{t-1} + \varphi_2 Y_{t-2} + \cdots + \varphi_p Y_{t-p} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

where Y_t represents the time series data; φ_1 to φ_p represent the AR parameters that can describe the relationship between the current value and the value of the past p time points; θ_1 to θ_q represent the MA parameters that are used to describe the relationship between the current value and the error of the past q time points; ε_t represents the error term at time t ; and c represents a constant term.

Indicator system and data sources

Urban agglomerations are complex systems formed by the combined effects of multidimensional elements, of which the social, economic and built environments are the most important dimensions. These elements interact in a complex manner. The social subsystem offers human and technological assistance to the economy, whereas the economic subsystem provides financial and product support to society^{34,35}. The social subsystem offers technical assistance to the built environment; however, social activities can also have a detrimental effect on the built environment. The built environment subsystem serves as the material foundation for social development³⁶. The economic subsystem provides the foundation for the built environment, yet economic activities can also result in environmental degradation. The built environment subsystem serves as a vital conduit and source of resources for economic activities³⁷ (Fig. 2).

(1) Social resilience (SR). SR indicators are used to assess the social capacity of urban agglomerations to remain stable, adapt and recover from various challenges and pressures³⁸. These indicators emphasize social cohesion, adaptability, resilience, and effective governance and leadership³⁹. Indicators are selected to allow comparisons across societies, regions or periods while reflecting the changing social environment⁴⁰.

(2) Economic resilience (ER). ER indicators are used to measure the economic development of urban agglomerations to maintain stability, rapidly recover and adapt in the face of various external shocks and internal challenges⁴¹. These indicators include the diversity of economic structure, market flexibility, and labour market adaptability⁴². The selection of indicators focuses on the robustness of the economy, its recovery and its ability to adapt and transform in the long run⁴³.

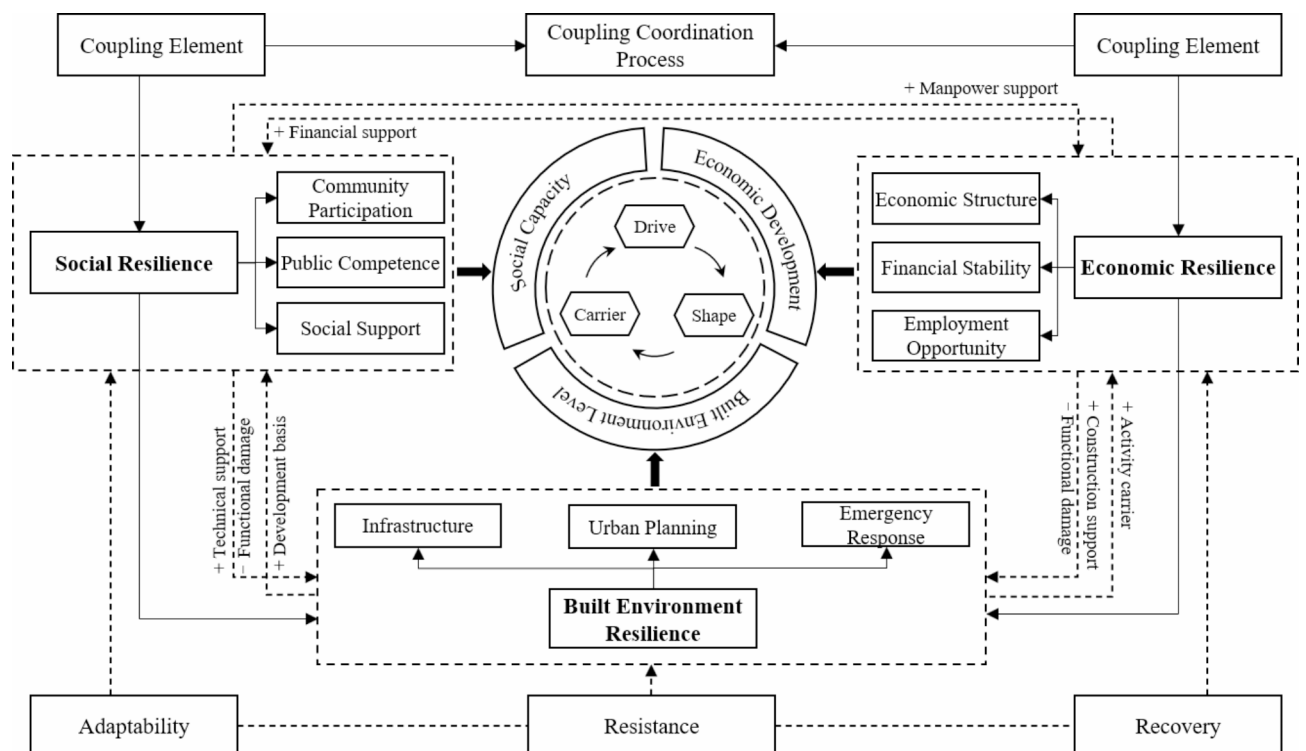


Fig. 2. Socio-economic-built environment resilience relationship model.

(3) Built environment resilience (BR). BR indicators are concerned with the stability, adaptability and recovery of urban and built environments in the face of natural disasters, climate change and socioeconomic shocks⁴⁴. These indicators include the durability of buildings, the reliability of infrastructure, the flexibility of public spaces, and the sustainability of the environment⁴⁵. In indicator selection, the emphasis is on the robustness of the structure, the redundancy of the system, and the scheduling of resources⁴⁶.

The screening indicators are mapped to the three capabilities that are involved in resilience: adaptability, resistance, and recovery. Adaptability emphasizes the ability of a system to adjust and optimize in the face of changes in the external environment. Resistance pertains to the capacity of a system to maintain stability in the face of unforeseen circumstances or disruptions in the immediate term. Recovery focuses on the ability to respond to emergencies and resume normalcy swiftly following a crisis. Therefore, the SR subsystem focuses on public service adaptation, social structural resistance and recovery staffing. The ER subsystem focuses on economic change adaptation, economic scale resistance and recovery stability. The BR subsystem focuses on engineered structural resistance and rescue and recovery facilities. Moreover, taking into account the quantifiable, representative and dynamic characteristics of the indicators, the indicator system was finally constructed under the three dimensions of SR, ER and BR, including a total of 18 quantifiable resilience indicators (Table 2). Among them, the unemployment rate, population ageing rate and urban-rural income ratio are negative indicators; i.e., the higher their values are, the lower the corresponding level of resilience. The remaining indicators are positive indicators; i.e., the higher their values are, the higher the corresponding level of resilience. The main source of data for the study is the China County Statistical Yearbook (Township) (2000–2020), published by the National Bureau of Statistics of China, which contains various types of social and economic data for 199 counties (cities and districts) in the BTH region. In addition, data related to the built environment are obtained from the Statistical Yearbook of Urban and Rural Construction (2000–2020), published by the Ministry of Housing and Urban-Rural Development of China.

Results
Spatial and temporal distributions of resilience in the BTH region

Temporal evolution characteristics of resilience
The resilience levels observed in different cities are reflective of the varying characteristics of changes that have occurred over the specified period. Cities such as Beijing and Xingtai displayed sustained growth in overall resilience. Tianjin and Handan, however, experienced an initial increase followed by a subsequent decrease in overall resilience. Cities such as Shijiazhuang and Tangshan showed a pattern of initial decline followed by an increase in overall resilience (Fig. 3-(a)). The SR levels in multiple cities exhibited a consistent downwards trend, notably including cities such as Langfang and Hengshui (Fig. 3-(b)). Throughout the research period, the ER levels in the majority of cities displayed a continuous upwards trend (Fig. 3-(c)). During the research period, cities such as Shijiazhuang and Tangshan experienced an increasing trend in BR levels. In contrast, some cities exhibited an initial increase in the rate of change, which was subsequently followed by a decrease. (Fig. 3-(d)).

Spatial distribution characteristics of resilience
With the use of the natural fracture method, the resilience scores of various counties in the BTH region were classified into five levels to identify the spatial distribution characteristics for each year. In 2000, the areas with

Dimension	Indicator	Resilience Capacity	Indicator Unit	Nature of Indicator	References	Data Source
Social (S)	S1: Urbanization rate	Resistance	%	+	47,48	China County Statistical Yearbook (Township) (2000–2020)
	S2: Unemployment rate	Recovery	%	-	8,34	
	S3: Population aging rate	Resistance	%	-	49	
	S4: Number of health technicians	Adaptability	Person	+	48,50	
	S5: Number of full-time teachers in primary and secondary schools	Adaptability	Person	+	51,52	
	S6: Number of various social welfare adoptive units	Adaptability	N	+	51,52	
Economic (E)	E1: Regional GDP	Resistance	million yuan	+	34,48	China County Statistical Yearbook (Township) (2000–2020)
	E2: Urban-rural income ratio	Recovery	-	-	53,54	
	E3: Share of secondary and tertiary industries in GDP	Adaptability	%	+	34,48	
	E4: Investment in fixed assets	Adaptability	million yuan	+	47,50	
	E5: Per capita disposable income of urban residents	Resistance	Yuan	+	47	
	E6: General budget revenue of local finance	Adaptability	million yuan	+	53,55	
Built Environment (B)	B1: Density of water supply pipes in built-up areas	Resistance	km/km ²	+	8,34	Statistical Yearbook of Urban and Rural Construction (2000–2020)
	B2: Density of drainage pipes in built-up areas	Resistance	km/km ²	+	8,34	
	B3: Green space per capita	Recovery	m ² /person	+	8,56	
	B4: Green land coverage ratio	Recovery	%	+	8,56	
	B5: Built-up land area ratio	Recovery	%	+	50,56	
	B6: Road area per capita	Recovery	m ² /person	+	34,48	

Table 2. Socio-economic-built environment resilience indicator system.

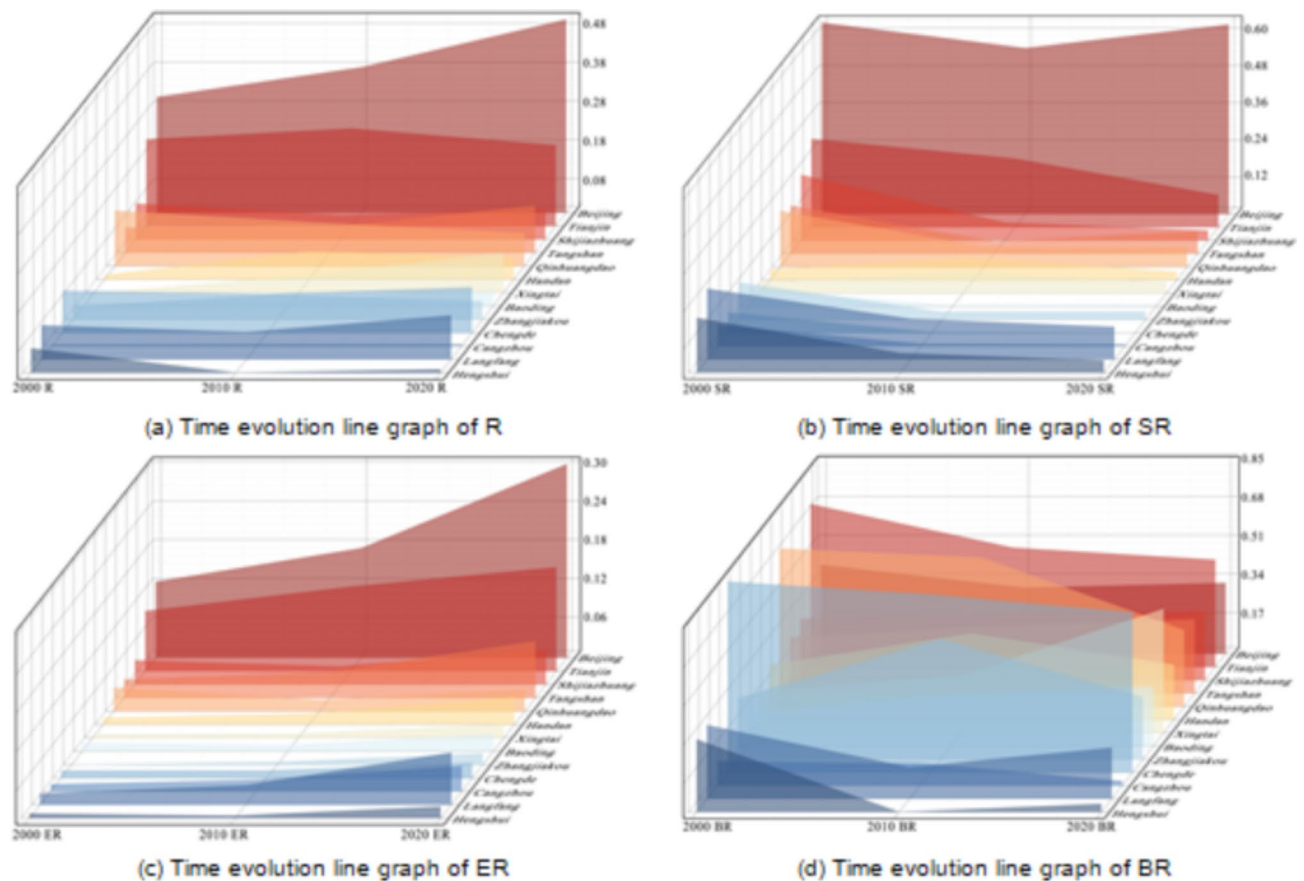


Fig. 3. Temporal changes in resilience levels by district and county, 2000–2020. The figure was created using Origin, version 2021 (<https://www.originlab.com>) <https://www.originlab.com>.

the highest resilience levels included Beijing's Dongcheng, Xicheng, Chaoyang, and Haidian districts, as well as the Tianjin Binhai New Area. The other areas of Beijing, Tianjin's Inner Six Districts, and Jizhou District followed closely. The northern part of the BTH region and the central-western part of Shijiazhuang also exhibited higher resilience. Relatively lower resilience levels appeared in the central-southern part of the BTH region (Fig. 4-(a)). By 2010, high-resilience areas were concentrated in the central areas of Beijing and Tianjin's Binhai New Area, whereas low-resilience areas were concentrated in the central-southern and northwest parts. Compared with 2000, this period showed a more pronounced trend of concentration in high- and low-resilience areas (Fig. 4-(b)). In 2020, the entire region of Beijing and Tianjin's Binhai New Area maintained high resilience levels, with the remaining areas of Tianjin also showing relatively high resilience. Additionally, certain areas in Tangshan, Handan, and Shijiazhuang exhibited high resilience. Moreover, low-resilience areas were concentrated in the central-southern and northwestern parts of China. Compared with those in 2010, high-resilience areas in Beijing became more concentrated (Fig. 4-(c)). Overall, the initial central and northern parts of the BTH region presented consistently high resilience levels. This trend strengthened in 2010 and 2020, highlighting the resilience levels of Beijing and slightly reflecting a Matthew effect.

Between 2000 and 2020, the overall trend indicates a decrease in the SR in the northwest, whereas the southern part of Shijiazhuang also exhibited a decreasing trend. The high-resilience areas ultimately converge in a continuous urban cluster comprising Beijing, Tianjin, and their eastern regions. This convergence reflects a transfer trend of the SR from west to east and from south to north (Fig. 4-(d), (e), (f)). In the economic dimension, the overall ER was initially relatively concentrated in Beijing and Tianjin, with high-value areas showing scattered distribution characteristics. High-value areas began to cluster and gradually spread to cities surrounding Beijing and Tianjin, demonstrating a distinct radiation effect. By 2020, high-value areas had formed a triangular cluster with these cities at their core, indicating the diffusion trend of the ER distribution (Fig. 4-(g), (h), (i)). In terms of the built environment, the overall characteristics indicate that the BR has certain geographical distribution features, with the mountainous areas in the northwest generally having greater geographical distribution features than the urban development areas in the southeast. Chengde consistently maintained the highest BR level. Additionally, high-value aggregation areas formed along mountainous and coastal areas, reflecting the advantages of the BR resulting from superior natural environmental conditions in these regions (Fig. 4-(j), (k), (l)).

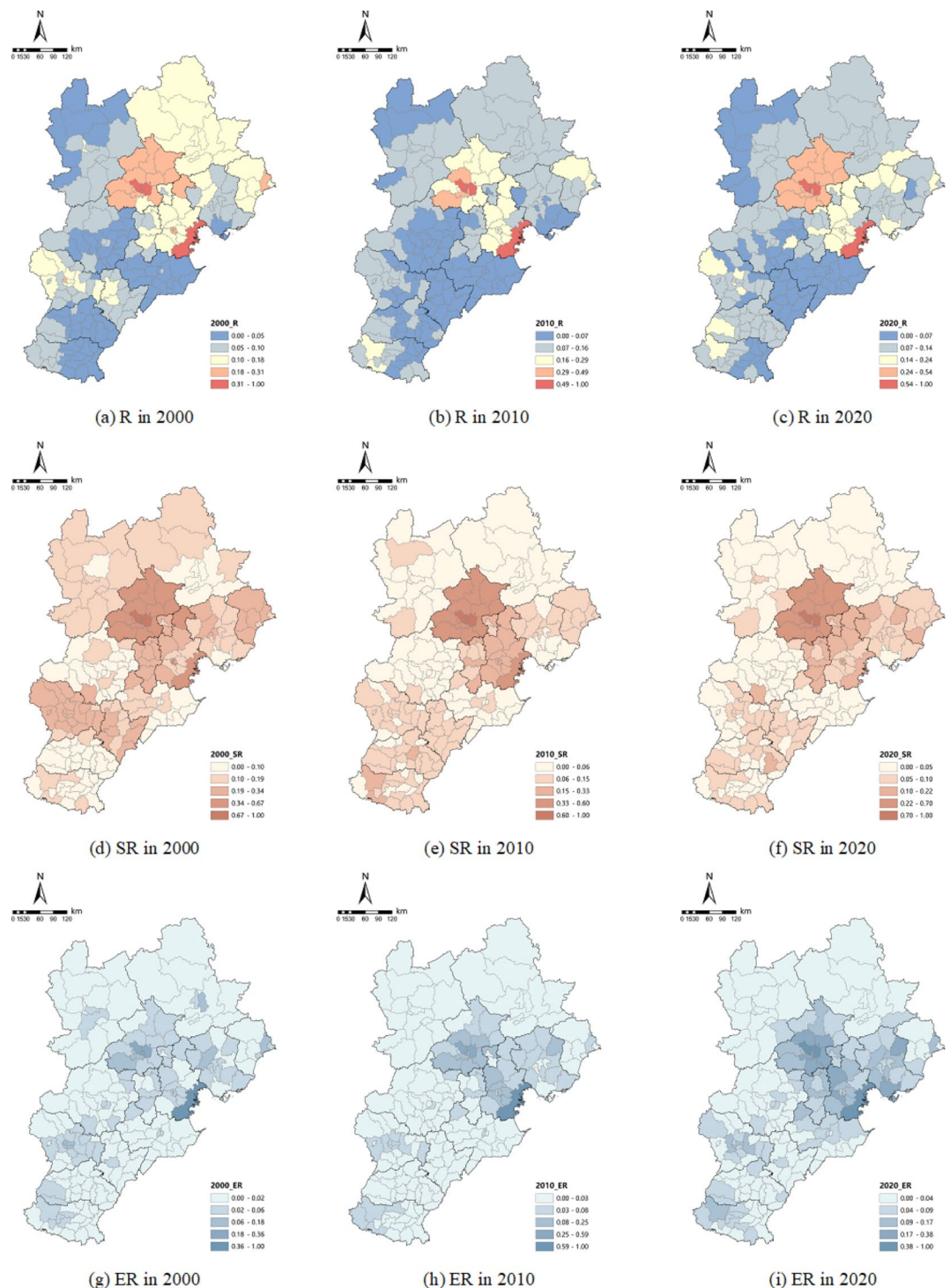


Fig. 4. Resilience distribution in 2000, 2010, and 2020. Produced based on the Ministry of Natural Resources Standard Map Service website, Standard Map No.GS (2016)1610, with no modifications to the boundaries of the base map. The map was created using ArcGIS, version 10.4 (<https://www.esri.com/en-us/arcgis>).

Spatial and temporal distributions of the CCD in the BTH region

Temporal evolution characteristics of the CCD

In 2000, studies of the CCD in the BTH regions S-E, S-B and E-B revealed distinct geographical variations. The high-CCD areas of S-E were concentrated in parts of Beijing and Tianjin, whereas the northwestern, central and southern tip regions faced imbalances. The northeastern cities performed better in terms of the S-B CCD, whereas the values in the southwestern regions were generally lower. The northeastern and western regions along the mountains were highly coordinated in E-B, but some areas, such as northwestern Zhangjiakou and southern

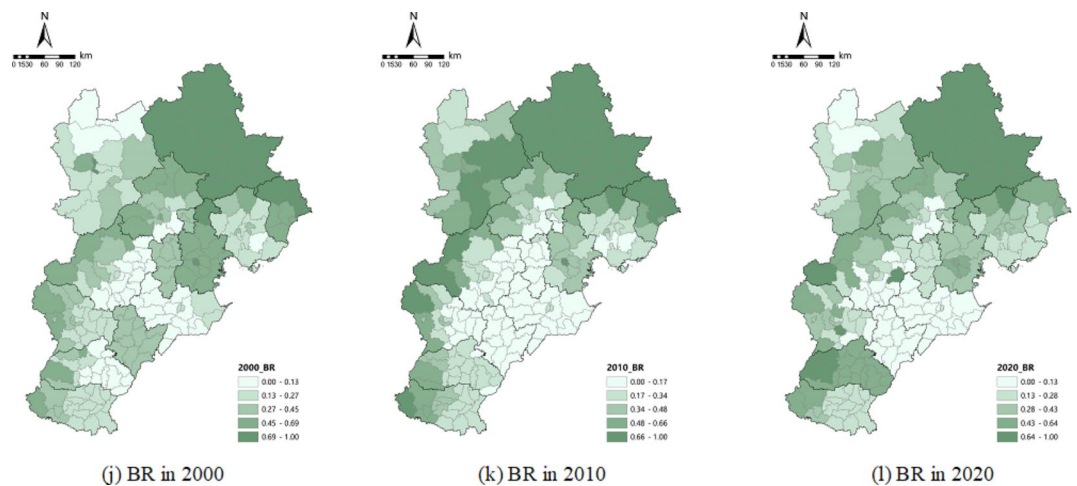


Figure 4. (continued)

Cangzhou, showed considerable imbalances. The Tianjin Binhai New Area performed best in 2000, with a high degree of social, economic and environmental coordination. While in the central area of Beijing, some areas of Tianjin and Qinhuangdao are essentially balanced, most of the remaining BTH area is moderately unbalanced, and some areas, such as northern Zhangjiakou and eastern Baoding, face substantial imbalances (Fig. 5).

In 2010, the CCD of S-E shows that high-coordination areas are more concentrated in Beijing and Tianjin, whereas most of the remaining areas are generally imbalanced or severely imbalanced. The CCD of S-B shows that northwest Beijing, west Tianjin, west Shijiazhuang and Handan, and Qinhuangdao are coordinated, but most of the areas are still imbalanced, and some areas, such as Hengshui, Cangzhou, and Baoding, are severely imbalanced in a small area. In the CCD of E-B, the Tianjin Binhai New Area has the highest degree of coordination, forming a band of high values from northeast to southwest, while the south-central part of the city forms a triangular zone of severe imbalance. While the core areas of Beijing and Tianjin maintain relative coordination, the proportion of severe imbalance in the BTH area increases, indicating an increasing trend of incoherence (Fig. 5).

In 2020, the S-E coordinated area of the BTH region increased compared with that in 2010, especially in the northeastern part of the region, reflecting improved coordinated socioeconomic development. There is little change in the CCD for S-B, with severe imbalances persisting in Cangzhou and Hengshui. The CCD for E-B shows an increase in the degree of coordination across the region, with the central region showing a trend from imbalance to basic coordination. Overall, the centres of Beijing and Tianjin are the regions with the highest degree of coordination, while the regions with severe imbalances are concentrated in Cangzhou and Hengshui, and most of the northern and western parts of BTH are in a state of general imbalance, revealing a new development balance and challenges (Fig. 5).

Spatial aggregation of CCDs

In 2000, the application of spatial clustering revealed distinct patterns within the BTH region. Areas exhibiting high-high clustering, notably Beijing, Tianjin, and Qinhuangdao, demonstrated strong social development, stable economies, and favourable environmental conditions. This finding reflects a high level of comprehensive development and balanced integration of social and economic factors. In contrast, regions exhibiting low-low clustering displayed comparatively lower CCDs, particularly in the western and southern regions, indicating a prevalence of challenges pertaining to social development, economic growth, and environmental quality. Furthermore, low-high clustering was identified in a restricted area of western Tangshan, indicating that localized factors contribute to the fragmentation of development. The analysis indicates that the central and northeastern regions had higher S-E-B CCDs in 2000, whereas the western and southern regions need further development efforts and policy support (Fig. 6a).

The spatial clustering of the BTH region in 2010 exhibited notable shifts. High-high clustering was predominantly observed in the central areas, particularly within the Beijing and Tianjin districts. Furthermore, an expanded cluster area was noted, reflecting growth in the western part of Beijing and the north-central region of Tianjin in comparison to the data from 2000. Conversely, the prevalence of low-low clustering increased in the southern region, encompassing the majority of Baoding, Cangzhou, and Hengshui. This increase was accompanied by a notable expansion of clustered areas, indicating a shift from a dispersed distribution to a concentration in the south-central part of the country. In general, the central region demonstrated a high level of coordinated socioeconomic–environmental development in 2010. Conversely, the southern regions need increased attention and development efforts (Fig. 6b).

In 2020, there were notable changes in spatial clustering in the BTH region, with high-high clustering concentrated in the central areas. The cluster size notably expanded to encompass all of Beijing, indicating a strong clustering effect. Low-low clustering persisted at the southern and northern edges, resembling the 2010 distribution but with a novel cluster emerging to the north of Zhangjiakou. This phenomenon reflects a

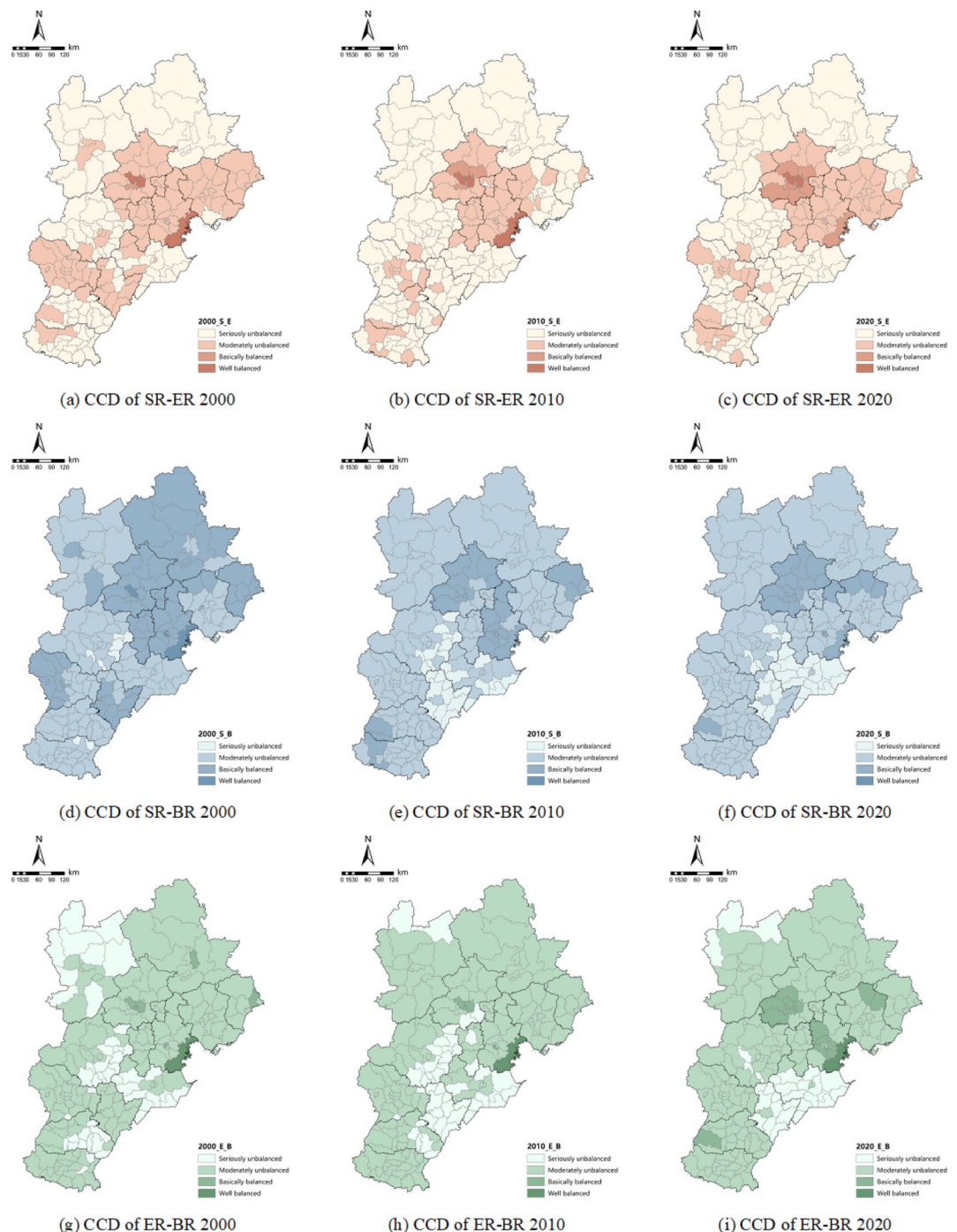


Fig. 5. CCD distribution in 2000, 2010, and 2020. Produced based on the Ministry of Natural Resources Standard Map Service website, Standard Map No.GS (2016)1610, with no modifications to the boundaries of the base map. The map was created using ArcGIS, version 10.4 (<https://www.esri.com/en-us/arcgis>).

transition from dispersion to aggregation and back. The occurrence of low-high clustering was less frequent in the southern part of Chengde, indicating that although these areas exhibited lower CCDs, their surrounding regions demonstrated greater performance. Overall, the central region demonstrated a high CCD, whereas a low CCD was more pronounced in the southern and northern regions. High-high Clustering consolidated towards the centre, whereas low-low Clustering fluctuated between the dispersed state and the concentrated state (Fig. 6c).

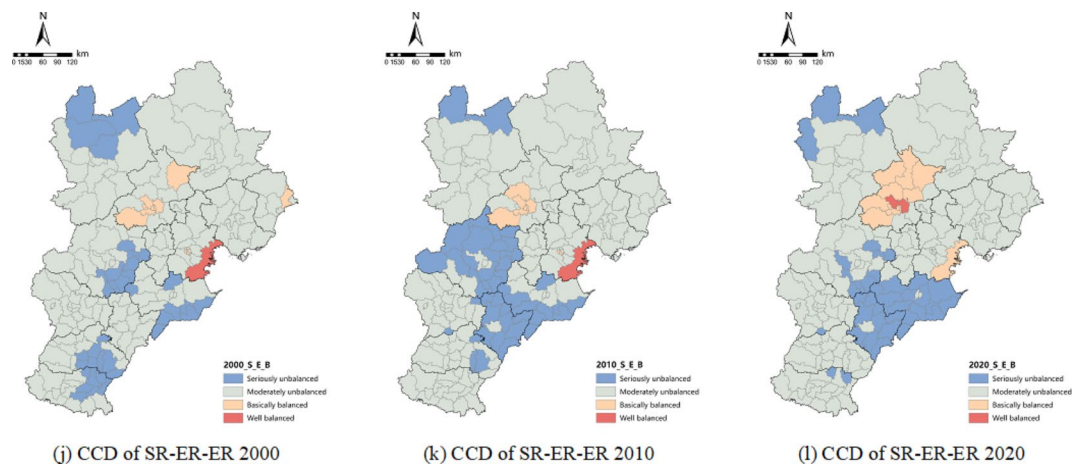


Figure 5. (continued)

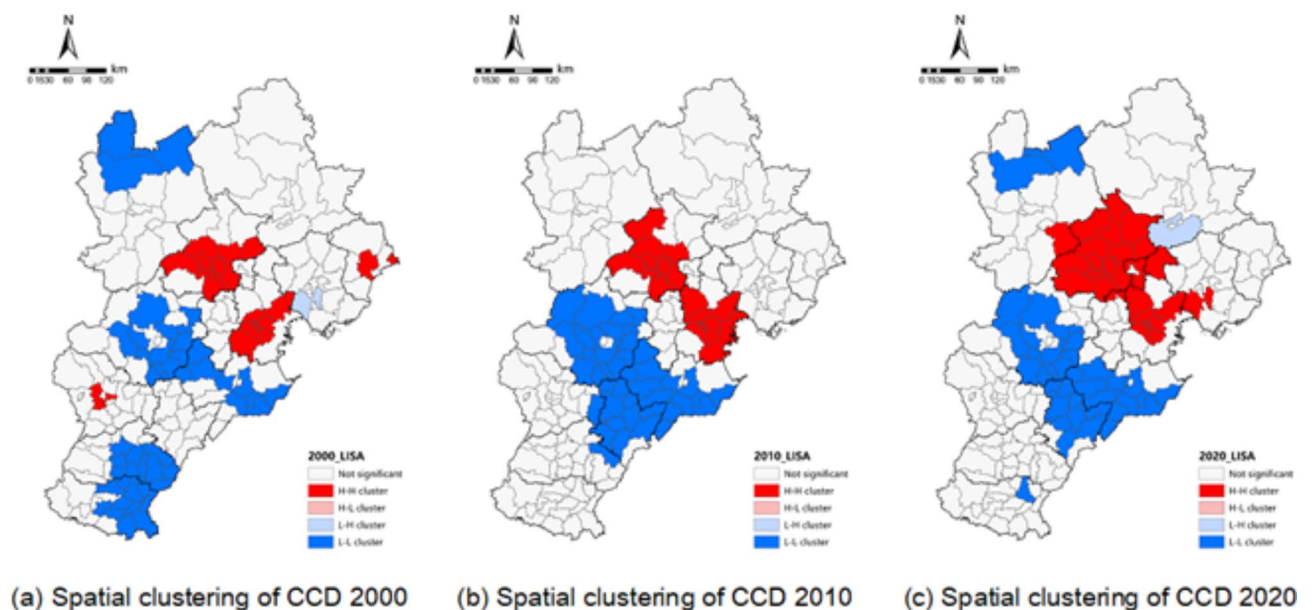


Fig. 6. Spatial clustering of CCD in 2000, 2010, 2020. Produced based on the Ministry of Natural Resources Standard Map Service website, Standard Map No.GS (2016)1610, with no modifications to the boundaries of the base map. The map was created using ArcGIS, version 10.4 (<https://www.esri.com/en-us/arcgis>).

Factor detection of CCD divergence in the BTH region

Results of the detection of dominant factors

The detection of the factors that may affect the CCD divergence provides the dominant factors contributing to the situation for each of the three years (Table 3). In 2000 and 2010, the factors with the first and second highest influence on the CCD divergence were investment in fixed assets (E4) and regional GDP (E1), and these two factors remained unchanged until 2020, but the order was reversed. In this period, economic factors, especially the economic aggregate status of the region as a whole, are the main factors causing interregional differences, reflecting that the BTH region as a whole is unbalanced in the development of economic aggregates. The E-type factors are always in the dominant position, and five of the six factors are always in the top seven positions.

Among the social factors, the number of full-time teachers in primary and secondary schools (S5) is the most influential, ranking sixth of all factors in 2000 and fourth in both 2010 and 2020. In addition to S5, the number of health technicians (S4) is the highest among the social factors, reflecting a certain dominance. These findings indicate that the social factors influencing the CCD divergence in the BTH region are mainly differences in education and medical conditions. The factor with the greatest influence on B-type land area was the built-up land area ratio (B5), which moved from seventh place in 2000 to eighth place in 2010 and 2020. Most of the factors in the built environment dimension have a limited impact on the CCD divergence, and the development of cities in these aspects is relatively balanced.

Year	Factor	S1	S2	S3	S4	S5	S6
2000	q	0.69843	0.50694	0.49036	0.93257	0.98606	0.36373
	p	0.00000	0.00000	0.00000	0.48249	0.02789	0.99988
2010	q	0.72289	0.72289	0.59503	0.96435	0.99781	0.49558
	p	0.00000	0.00000	0.00000	0.00000	0.00000	0.94677
2020	q	0.79712	0.79712	0.63597	0.96786	0.99451	0.66255
	p	0.000	0.000	0.000	0.01772	0.00000	0.78743
Year	Factor	E1	E2	E3	E4	E5	E6
2000	q	0.99862	0.93855	0.99440	0.99909	0.99592	0.99632
	p	0.00000	0.40842	0.00000	0.00000	0.00000	0.00000
2010	q	0.99902	0.92751	0.98858	0.99924	0.99560	0.99826
	p	0.00000	0.01555	0.00245	0.00000	0.00000	0.00000
2020	q	0.72289	0.72289	0.72289	0.49557	0.95510	0.72289
	p	0.00000	0.00000	0.00000	1.00000	0.88116	0.00000
Year	Factor	B1	B2	B3	B4	B5	B6
2000	q	0.69843	0.69843	0.69843	0.62711	0.97717	0.69843
	p	0.00000	0.00000	0.00000	1.00000	0.46250	0.00000
2010	q	0.72289	0.72289	0.72289	0.49557	0.95510	0.72289
	p	0.00000	0.00000	0.00000	1.00000	0.88116	0.00000
2020	q	0.79712	0.79712	0.79712	0.82351	0.93487	0.79712
	p	0.00000	0.00000	0.00000	0.99999	0.92758	0.00000

Table 3. CCD drive factor detection results by year.

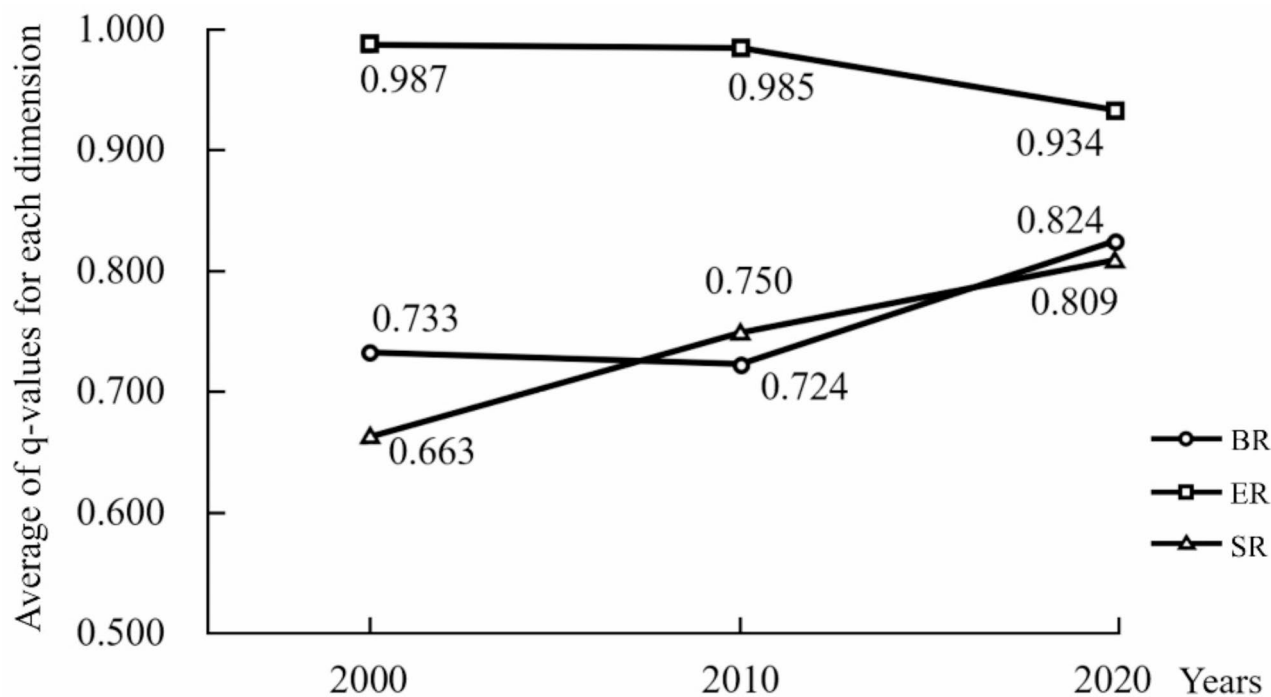


Fig. 7. Main dimensions affecting CCD over time.

Changes in dominant factors over time

To analyse the changes in the main dimensions affecting the CCD, the mean value of the factor influence level q under the three dimensions was calculated (Fig. 7). The economic factor was the most dominant factor influencing the CCD, which was much greater than the social and built environment. However, its dominant role shows a decreasing trend over the period 2000–2020, with a particularly significant decrease in the period 2010–2020, reflecting an increase in the balance of economic development in the BTH region. The social factor had a relatively small influence on the CCD in approximately 2000 but showed an increasing trend, and in 2010,

its influence exceeded that of the built environment factor. By 2020, its influence was slightly lower than that of the built environment but still greater its influence in 2000. The level of social development plays an increasingly important role in the equilibrium of the BTH region. The influence of the built environment factor on the CCD was in the middle of the three dimensions in approximately 2000 and was slightly lower than that of the social factor in 2010, but its influence increased significantly from 2010 to 2020, returning to second place in terms of influence and increasing significantly in absolute terms, reflecting the increased importance of this dimension.

Future projections for the BTH region CCD

Projections of the future spatial distribution of the CCD

The prediction of the future spatial distribution of the CCD is based on the natural development scenario, i.e., the predicted pattern without the influence of political intervention, economic transformation, etc. (Fig. 8). A comparison of the changes in the CCD in each decade from 2000 to 2050 reveals that the CCD of S-E-B in the region evolves significantly, and both the well-balanced region and the highly unbalanced region show a reciprocal process from dispersed to concentrated and then from concentrated to dispersed. The projections for 2030 and 2050 show an optimistic trend; i.e., more regions have reached a state of basic balance, and the regions with severe imbalance have decreased from the status quo. The projections for 2030 show an increase in the CCD in the central region, especially in the districts and counties of Baoding and Cangzhou, whereas the projections for 2050 show that a larger proportion of the region has moved towards basic or good balance transformation. In the future, areas facing coherence challenges are likely to be concentrated in a small number of districts and counties in northern Zhangjiakou, southern Cangzhou and central Baoding, requiring more targeted measures.

Future CCD spatial clustering prediction

Compared with the spatial clustering of the CCD in each decade from 2000 to 2050, the area of H-H clustering shows an increasing and then a decreasing trend, whereas the area of L-L clustering shows an increasing and then a stabilizing trend. The projections for 2030 show that the H-H clustering area is expected to further consolidate in the centres of Beijing and Tianjin, forming a more stable area with high CCD values. In contrast, the L-L clustering area is concentrated in parts of Baoding, Cangzhou and Hengshui in southern China (Fig. 9). By 2050, projections indicate that this concentration trend may slow or even partially reverse. Although the central area still has a high CCD, the extent of H-H clustering decreases. This finding suggests that after a long period of concentrated development, the peripheral areas may begin to gain more development opportunities or that the development of the central area may reach a saturation point, leading to a shift in development priorities (Fig. 9).

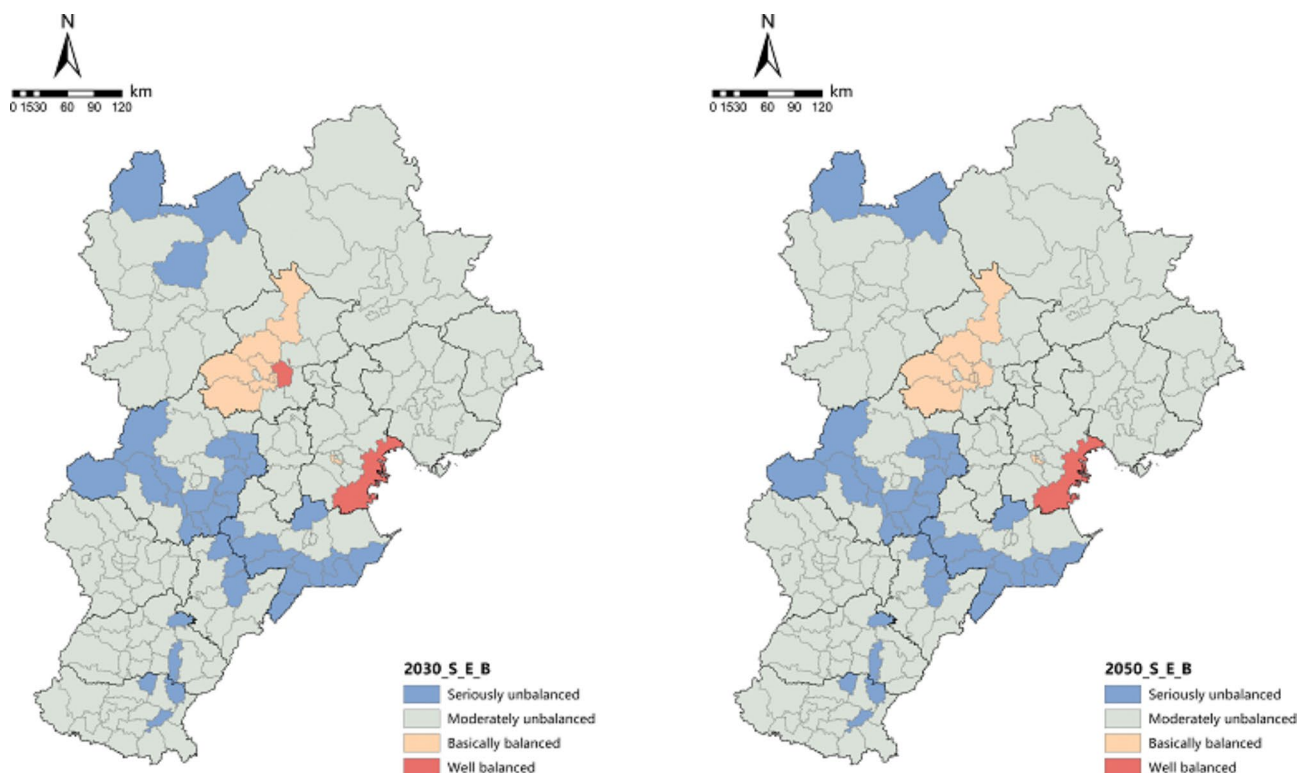


Fig. 8. Prediction of CCD distribution in 2030, 2050.

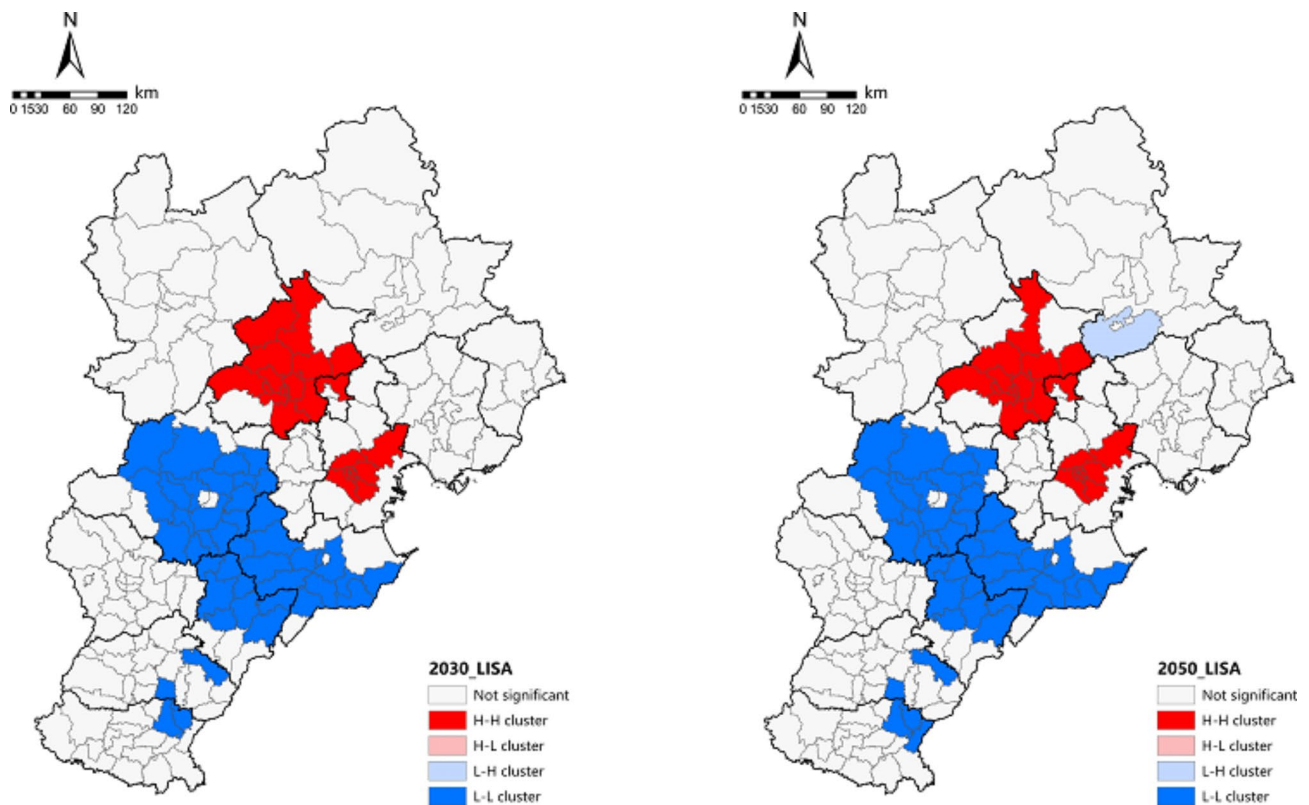


Fig. 9. CCD spatial clustering projections for 2030, 2050. Produced based on the Ministry of Natural Resources Standard Map Service website, Standard Map No.GS (2016)1610, with no modifications to the boundaries of the base map. The map was created using ArcGIS, version 10.4 (<https://www.esri.com/en-us/arcgis>).

Discussion

Spatiotemporal differentiation of resilience in the BTH region

Analysis of the temporal evolution of resilience

The results reveal that the evolution of urban resilience in the BTH region has manifested differently across various types of cities. The initial group of cities, including Beijing, Xingtai, Baoding, and Cangzhou, demonstrated a sustained enhancement in their urban resilience. In combination with the policies that have been in place over the specified period and the findings of existing studies, these cities share several common characteristics. They are regional centres or key development cities, which affords them greater policy support and investment, more stable economic growth, and diversified economic structures. This facilitates the improvement of infrastructure and social services⁵⁷. In contrast, Tianjin, Handan and Zhangjiakou exhibited fluctuating resilience. These cities have experienced a decrease in resource-dependent industries (Handan, Zhangjiakou) or have undergone significant restructuring of their industrial structure (Tianjin) regarding economic and social aspects, with associated pressures⁵⁸. Furthermore, the concurrent impact of environmental concerns and shifts in the external market has also influenced these cities⁵⁹.

The overall resilience of some cities, including Shijiazhuang, Tangshan, Chengde, Langfang, and Hengshui, has decreased and subsequently increased. Studies have demonstrated that the majority of these cities underwent significant economic and social reforms over two decades, subsequently identifying new growth points following the initial decrease. These cities have undergone industrial transformation and upgrading, developing new industries, including smart manufacturing, tourism and recreation. This change has contributed to economic, social and environmental development. Notably, the overall resilience of Qinhuangdao city demonstrated a persistent decrease from 2000 to 2020, which may reinforce the hypothesis that industrial diversity is a contributing factor for urban resilience. A diversified industrial structure has been demonstrated to be positively correlated with the level of regional resilience⁶⁰. Conversely, a homogenized society may be more susceptible to external shocks.

Analysis of the spatial evolution of resilience

The results revealed a distinct differentiation of aggregation among cities, forming a high-resilience zone centred on Beijing and Tianjin. Beijing, as the capital of China, has the administrative centre of the country and important governmental institutions; Tianjin, as one city directly under the central government, also enjoys special administrative and economic policies and therefore has an absolute advantage in the development of society, the economy and the built environment⁶¹. In the context of social resilience, high-quality areas of

Shijiazhuang, Langfang, Tangshan, and Qinhuangdao and a low-quality area of Baoding are formed. Most of the cities are located in the Beijing and Tianjin metropolitan areas and have unique geographical and social development advantages. Shijiazhuang, as the capital of Hebei Province, performs important administrative and service functions and is rich in public service resources such as medical care and education. In terms of economic resilience, the high-value areas are concentrated in Beijing, Tianjin, Shijiazhuang and Tangshan. These are key cities in the BTH region, and some cities are important on the development axis of BTH⁶². Zhangjiakou and Chengde in the northern part of the region are experiencing slower economic development, which is constrained by environmental and resource conditions⁶³. In terms of the built environment, most of the districts and counties along the mountains in the northwestern part of the BTH region have a high BR, with high green cover and a superior ecological environment, forming a natural barrier, which is in accordance with the results of previous studies in this field⁶⁴. In comparison with existing studies, this study identifies more detailed characteristics of the spatial distribution of resilience by utilizing counties as the fundamental unit of analysis⁵⁹. In other words, the distribution of resilience levels is not entirely divided by city. Even counties within the same city exhibit considerable variation in resilience levels. High-value areas are concentrated in the core four districts of Beijing and the Binhai New Area of Tianjin, whereas low-value areas are distributed in the northwestern counties of Chengde and the southeastern counties of BTH.

Spatiotemporal differentiation of CCD in the BTH region

Analysis of the temporal evolution of the CCD

Following an analysis of the changes in the degree of coordination of the socioeconomic–environmental CCD in 2000, 2010 and 2020, the following general evolutionary trends are summarized: (1) Enhanced development of the central region. From 2000 to 2020, the CCDs of central Beijing, Tianjin and their peripheral regions continued to strengthen, demonstrating the increasing development coordination of these areas and their integration of social, economic and environmental aspects. (2) Divergent development in peripheral areas. Some districts and counties in Baoding, Cangzhou and Hengshui in southern China have lower CCDs at all three time points, a trend that becomes more pronounced in 2020, suggesting that the development of these regions may need more attention and resources. (3) Changes in the balance of regional development. In 2000, regions were more dispersed in terms of CCD, but by 2020, both high- and low-CCD regions appeared to be more concentrated, which may indicate that interregional development disparities increased over the study period. (4) Dynamics of change. Over time, interregional differences in coordination show some dynamism, with the continued strengthening of the central regions contrasting with the lagging development of the peripheral regions, possibly reflecting a concentration or shift in policy drivers, economic investment or socioenvironmental factors.

Compared with existing studies, this study contributes by projecting near- and long-term future CCD levels on the basis of multiyear status quo CCD evaluations^{21,24}. In general, the CCD levels in the BTH region are expected to increase after 2030 and 2050. This results in more counties reaching the basic equilibrium state and fewer counties exhibiting severe imbalances. Although the clustering results for the CCD in 2030 indicate an increase in the aggregation of high values in the core counties of Beijing and Tianjin, this trend is expected to be mitigated by 2050. This finding suggests that the regional imbalance may be resolved to a certain extent in the long term²².

Analysis of the spatial evolution of CCD

A summary of the study results indicates that the spatial distribution of CCDs in the BTH region exhibited distinct divergent characteristics, forming four major clusters. The initial cluster is the S–E–B aggregation area. The aforementioned clusters are concentrated in the districts and counties of the cities of Beijing and Tianjin, which have undergone coordinated development across three dimensions: society, the economy and the built environment. This has resulted in a high level of resilient development. Previous studies have demonstrated that the BTH region as a whole presents a dual-core driven situation centred on Beijing and Tianjin⁶⁵. The second area is the S–E aggregation area. The area in question is concentrated in the districts and counties of Tangshan, Langfang and Shijiazhuang. Notably, Tangshan and Langfang are located within the central zone of the BTH region, and as a result of the growth of the Beijing and Tianjin conurbations, the number of medical, educational and industrial establishments in the area has significantly increased⁶⁶.

The third area of investigation is that of S–B aggregation. The area is concentrated in the districts and counties of the cities of Chengde and Qinhuangdao. The area is distinguished by a superior ecological environment, proximity to the sea or mountains, and the prominence of tourism as a primary industry³¹, which drives the advancement and enhancement of social services. The built environment functions as a catalyst for the evolution of social services. The final category is the E–B aggregation area. The economies and built environments of these areas have developed in a coordinated manner, with the economies concentrated in districts and counties along the east coast and the built environments reflected in the mountainous terrain to the west⁶⁷. This pattern is indicative of the manner in which the ecological environment within these areas is conducive to the promotion of economic benefits.

Guidance for future development

By integrating the characteristics of spatiotemporal divergence of resilience, the analysis of CCD spatiotemporal divergence characteristics and the detection results of dominant factors, the cities in the BTH region can be classified into types and categorized to guide their future improvement priorities:

(1) The S–E–B aggregation area serves as a growth pole, and its further enhancement of societal, economic, and built environment resilience is achievable through the optimization of infrastructure and urban planning. In particular, through regional synergetic development strategies such as regional economic integration, the area

can facilitate resource sharing, reduce administrative barriers and realize complementary development between cities, thereby promoting and radiating the sustainable development of neighbouring cities.

(2) Despite notable advancements in its social and economic development, the built environment in the S-E aggregation area still needs improvement. The results of the study indicate that the most pivotal factor within the built environment dimension is the ratio of developed land. Therefore, the implementation of reasonable control over the proportion of developed land represents a crucial aspect to be considered in the development of this specific type of city. Upgrades to the built environment may be achieved through the implementation of environmental management and greening initiatives, as well as infrastructure enhancements. This encompasses industrial transformation and upgrading, the reinforcement of industrial pollution control measures, and the enhancement of urban green space coverage, thereby ensuring an improvement in a city's resilience.

(3) The S-B aggregation area is distinguished by more advanced development of social and built environments. The results of the dominant factor detection indicate that economic factors, particularly the total economic status of the region, are the most significant dimensions influencing the level of CCD in the region. Therefore, this type of city can achieve substantial progress through economic advancement. The enhancement of economic strength may be achieved through the development of high-quality tourism, the attraction of high-tech and green industries, the improvement of the quality of the labour force, and the improvement of transport and communication infrastructure. Favourable social and environmental conditions will be fully exploited and transformed into economic advantages for sustainable development.

(4) The E-B aggregation area has a high level of economic and environmental development. The principal factors within the social dimension are those pertaining to the levels of education and health care. Therefore, the level of social development may be enhanced through increased investment in education and health care, improved talent training and medical services, and the reinforcement of community and cultural services. These measures enhance the well-being of the population, reinforce social cohesion and establish a robust social foundation for economic growth.

(5) The south-central area of the BTH region is the most vulnerable zone. First, there is a need to reinforce infrastructure and enhance transport networks to improve urban connectivity and attractiveness. Second, the industrial sector should undergo a process of modernization and expansion, with the introduction of new industries and the cultivation of existing industries. Concurrently, the provision of public services, including education, health care and community services, should be enhanced to reinforce social resilience and facilitate balanced regional development.

Limitations and prospects

To explore the relationships among the subsystems of urban agglomerations and identify the key elements affecting the development of urban agglomerations, this study analyses the level of social, economic and built environment resilience and the related coordination relationships in the BTH region. However, this study has several limitations due to various subjective and objective conditions. First, obtaining highly accurate data is difficult. Therefore, the indicators selected for the study are limited to the data that can be accessed in the Statistical Yearbook. However, the social and economic systems of urban agglomerations are highly complex, and many equally important factors need to be included in future studies. Consideration could be given to field research, simulated observations and other means of obtaining more data to complement and improve the existing research system.

Conclusion

By constructing an S-E-B resilience indicator system, this study quantitatively measured the coupled and coordinated relationships among the resilience subsystems of urban agglomerations. By evaluating the resilience level of the BTH region and predicting the future resilience level, the following conclusions can be drawn:

(1) In terms of spatial distribution, regions with low resilience levels are significantly correlated with regions with low CCD levels, such as Hengshui city and southern Cangzhou city. This phenomenon is caused mainly by the notion that high-CCD regions must simultaneously satisfy the condition that all subsystems have high resilience values. Therefore, a low level of resilience in one of the subsystems can cause an imbalance in the whole system.

(2) During the study period, the resilience development level of cities within the BTH region was influenced mainly by economic factors, and there were significant differences in economic levels between cities. However, from 2000 to 2020, the differences in ER narrowed. In particular, there are large differences among cities in the three subfactors of government revenue, health and education, which need to be targeted for improvement in the future.

(3) During the study period from 2000 to 2020, the spatial distribution of the CCD in the BTH region underwent a dynamic process of change from dispersed to concentrated to dispersed. Before 2020, the gap between Beijing and Tianjin as the regional duo and the cities in Hebei will gradually increase, but it is predicted that this gap will narrow by 2030 and 2050. Therefore, of the studied cities should formulate and implement targeted future development strategies on the basis of their current characteristics.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Received: 13 August 2024; Accepted: 15 October 2024

Published online: 02 November 2024

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Author contributions

Hanyan Li and Qiao Wang wrote the main manuscript text. Xinyu Zang provided data and supervision. Tiancheng Gao, Haozhuo Gu prepared Figs. 4, 5 and 6. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-024-76653-x>.

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